EXPLORATORY DATA ANALYSIS REPORT

Week 1 Task: Online Retail Customer Purchase Behavior Analysis

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Data Analysis & Business Understanding

Week 1 Task — Exploratory Data Analysis Report

Dataset: Online Retail Transaction Data (UK-based online retailer)

Prepared as part of the Data Analysis & Business Understanding internship task.

Objective

A retail company wants to understand customer purchase behavior before investing in a recommendation system. This report explores the provided transaction dataset, identifies patterns and data quality issues, and presents a *complete Exploratory Data Analysis (EDA) covering who the customers are, which products sell the most, and which countries generate the most revenue.*

Dataset Note

The task references the Online Retail II dataset (2009-2011, UCI). This report uses the original Online Retail dataset (01/12/2010 - 09/12/2011), which shares an identical schema (Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country) .

1. Dataset Overview

The dataset was loaded and inspected for shape, columns, and data types before any analysis was performed using **pandas** library.

Load the dataset

```
# Load the dataset
df = pd.read_excel(r"D:\ITSimplera\online_retail_II.xlsx")
# Display first five rows
df.head()
```

[5] ✓ 1m 7.5s

```
   Invoice StockCode                          Description  Quantity  \
0   489434     85048  15CM CHRISTMAS GLASS BALL 20 LIGHTS        12
1   489434    79323P                   PINK CHERRY LIGHTS        12
2   489434    79323W                  WHITE CHERRY LIGHTS        12
3   489434     22041         RECORD FRAME 7" SINGLE SIZE         48
4   489434     21232       STRAWBERRY CERAMIC TRINKET BOX        24

          InvoiceDate  Price  Customer ID         Country
0 2009-12-01 07:45:00   6.95      13085.0  United Kingdom
1 2009-12-01 07:45:00   6.75      13085.0  United Kingdom
2 2009-12-01 07:45:00   6.75      13085.0  United Kingdom
3 2009-12-01 07:45:00   2.10      13085.0  United Kingdom
4 2009-12-01 07:45:00   1.25      13085.0  United Kingdom
```

Displaying basic information

- **Shape (df.shape)**
- **Column name (df.column)**
- **Data types (df.dtypes)**
- **General Information (df.info())**

```
# Display the number of rows and columns
print("Shape of Dataset:", df.shape)

# Display column names
print("\nColumns:")
print(df.columns)

# Display data types
print("\nData Types:")
print(df.dtypes)

# General information about the dataset
df.info()
```

```
Shape of Dataset: (525461, 8)

Columns:
Index(['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'Price', 'Customer ID', 'Country'],
      dtype='object')

Data Types:
Invoice                object
StockCode              object
Description            object
Quantity                int64
InvoiceDate    datetime64[ns]
Price                 float64
Customer ID           float64
Country                object
dtype: object
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525461 entries, 0 to 525460
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   Invoice      525461 non-null  object
 1   StockCode    525461 non-null  object
 2   Description  522533 non-null  object
...
 6   Customer ID  417534 non-null  float64
 7   Country      525461 non-null  object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 32.1+ MB
```

2. Missing Values & Duplicate Rows

Per task requirements, no rows were removed at this stage — this section is observation only.

Missing values checked using **df.isnull().sum() and** duplicates using **df.duplicated().sum()**

Column	Missing Count	Missing %
CustomerID	107927	24.93%
Description	2928	0.27%

Check Missing Values

```python
# Count missing values in each column
missing_values = df.isnull().sum()

print("Missing Values:")
print(missing_values)
```

[7] ✓ 0.1s

```
Missing Values:
Invoice            0
StockCode          0
Description     2928
Quantity           0
InvoiceDate        0
Price              0
Customer ID   107927
Country            0
dtype: int64
```

Check Duplicate Rows

```python
# Count duplicate records
duplicates = df.duplicated().sum()

print("Duplicate Rows:", duplicates)
```

8] ✓ 0.5s

```
Duplicate Rows: 6865
```

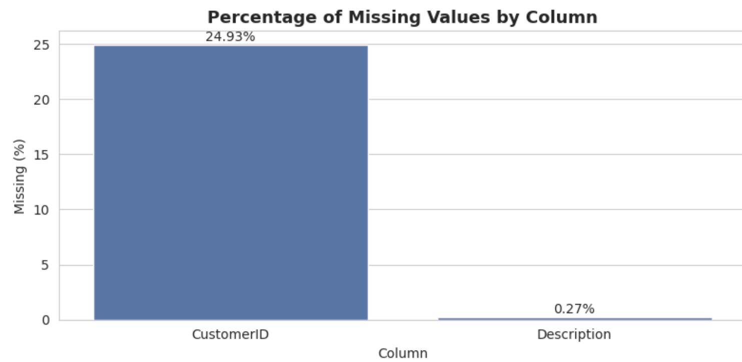

Figure 1: Percentage of missing values by column.

A total of 6,865 fully duplicated rows were found (0.97% of the dataset), likely a mix of genuine repeated line items on the same invoice and possible data entry duplication.

3. Top 10 Best-Selling Products

Visualized using matplotlib

By Quantity Sold

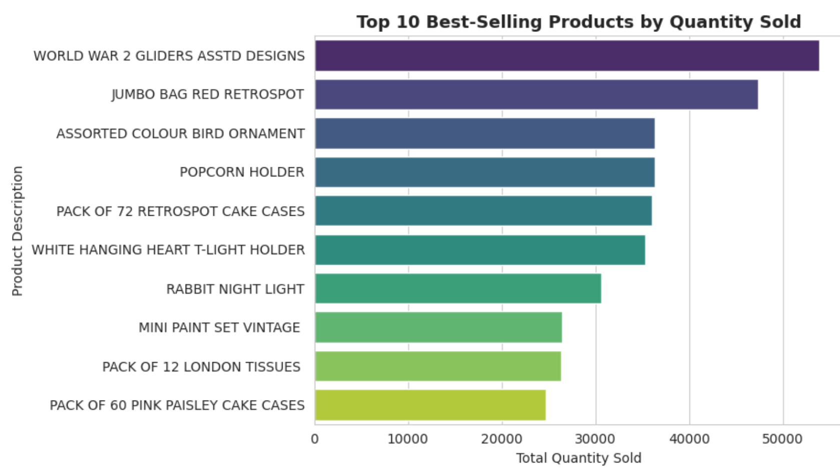

Figure 2: Top 10 products ranked by total quantity sold.

Rank	Product	Quantity Sold
1	World War 2 Gliders Asstd Designs	53,847
2	Jumbo Bag Red Retrospot	47,363
3	Assorted Colour Bird Ornament	36,381
4	Popcorn Holder	36,334
5	Pack of 72 Retrospot Cake Cases	36,039
6	White Hanging Heart T-Light Holder	35,317

Rank	Product	Quantity Sold
7	Rabbit Night Light	30,680
8	Mini Paint Set Vintage	26,437
9	Pack of 12 London Tissues	26,315
10	Pack of 60 Pink Paisley Cake Cases	24,753

By Revenue

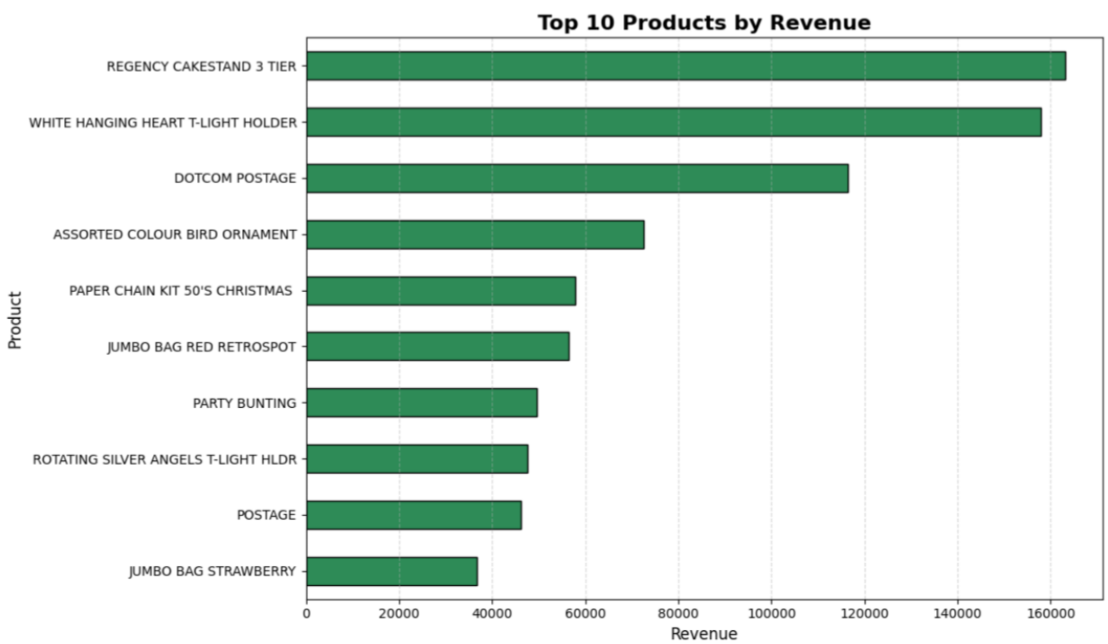

Figure 3: Top 10 products ranked by total revenue generated.

Rank	Product	Revenue (£)
1	Dotcom Postage	206,245.48
2	Regency Cakestand 3 Tier	164,762.19
3	White Hanging Heart T-Light Holder	99,668.47
4	Party Bunting	98,302.98
5	Jumbo Bag Red Retrospot	92,356.03
6	Rabbit Night Light	66,756.59
7	Postage	66,230.64
8	Paper Chain Kit 50's Christmas	63,791.94
9	Assorted Colour Bird Ornament	58,959.73
10	Chilli Lights	53,768.06

4. Sales Performance by Country

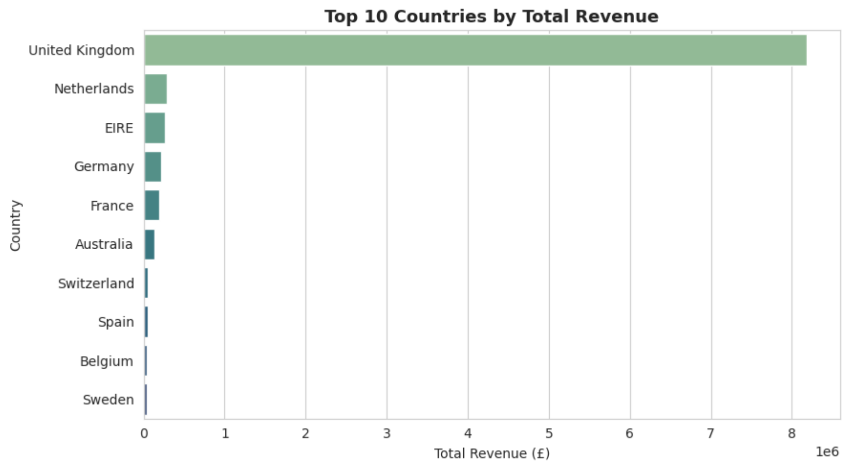

Figure 4: Top 10 countries ranked by total revenue.

Rank	Country	Revenue (£)	Orders
1	United Kingdom	8,187,806.36	23,494
2	Netherlands	284,661.54	101
3	Eire (Ireland)	263,276.82	360
4	Germany	221,698.21	603
5	France	197,403.90	461
6	Australia	137,077.27	69
7	Switzerland	56,385.35	74
8	Spain	54,774.58	105
9	Belgium	40,910.96	119
10	Sweden	36,595.91	46

The United Kingdom alone accounts for approximately 84.0% of total revenue, confirming the dataset's strong domestic concentration.

5. Revenue Over Time (Monthly Trend)

Figure 5: Total monthly revenue from December 2010 to December 2011.

Revenue rises steadily from August through November 2011, peaking in November at approximately £1.46M — consistent with pre-holiday/Christmas purchasing — before dropping sharply in the partial month of December (data ends 9 Dec 2011).

6. Correlation Heatmap

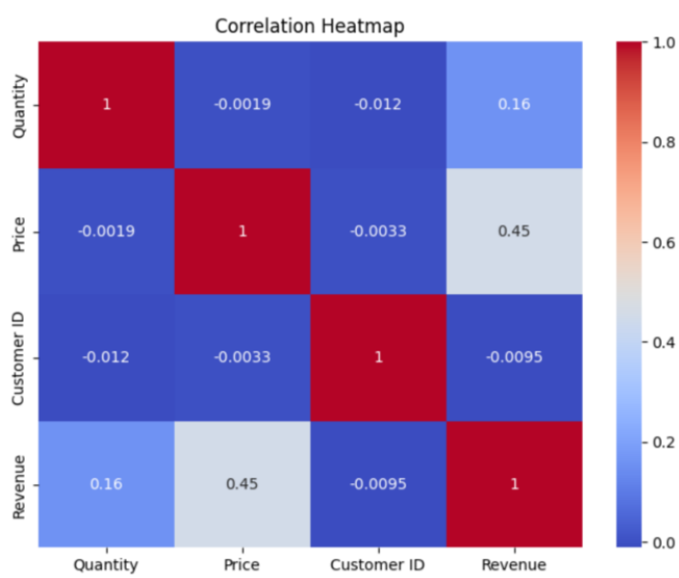

Figure 6: Correlation heatmap of Quantity, Price, Revenue, and CustomerID.

The correlation analysis shows that Price has a moderate positive relationship with Revenue (r = 0.45), while Quantity has only a weak positive correlation (r = 0.16). The relationships between Quantity and

Price, as well as Customer ID with all other variables, are negligible, indicating little or no linear association.

7. Outlier Detection (Box Plots)

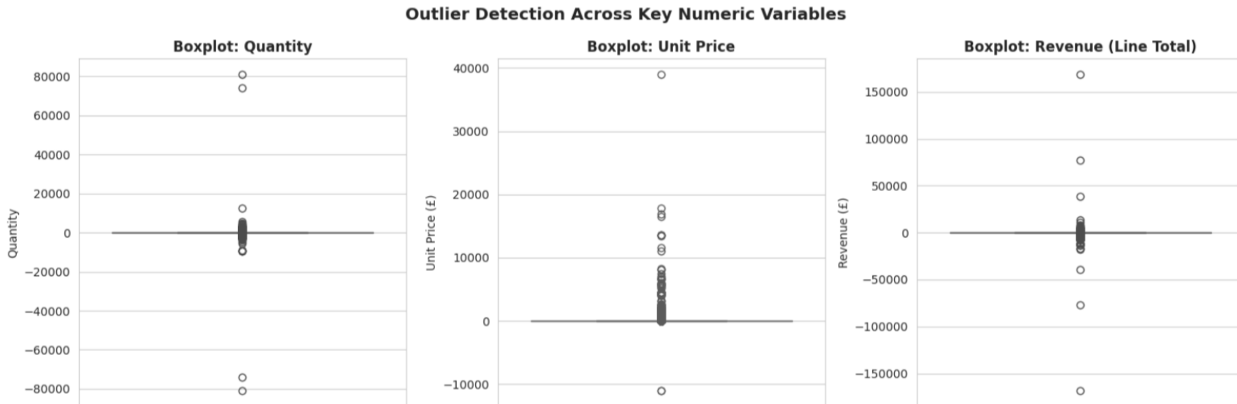

Figure 7: Box plots for Quantity, Unit Price, and Revenue, showing the spread of outliers.

Variable	Outliers (IQR method)	% of Dataset
Quantity	57870	11.01%
Price	35,273	6.71%
Revenue	45,872	8.73%

Outliers value

```
def iqr_outlier_count(series):
    q1, q3 = series.quantile([0.25, 0.75])
    iqr = q3 - q1
    lower, upper = q1 - 1.5*iqr, q3 + 1.5*iqr
    return ((series < lower) | (series > upper)).sum()

for col in ['Quantity', 'Price', 'Revenue']:
    n_out = iqr_outlier_count(df[col])
    print(f"{col}: {n_out} outliers ({n_out/len(df)*100:.2f}% of rows) using IQR method")
```

```
Quantity: 57870 outliers (11.01% of rows) using IQR method
Price: 35273 outliers (6.71% of rows) using IQR method
Revenue: 45872 outliers (8.73% of rows) using IQR method
```

All three numeric variables show substantial outlier proportions, largely attributable to bulk/wholesale orders and order cancellations (negative quantities). No outliers were removed, consistent with the observation-only scope of this task.

8. Customer Overview

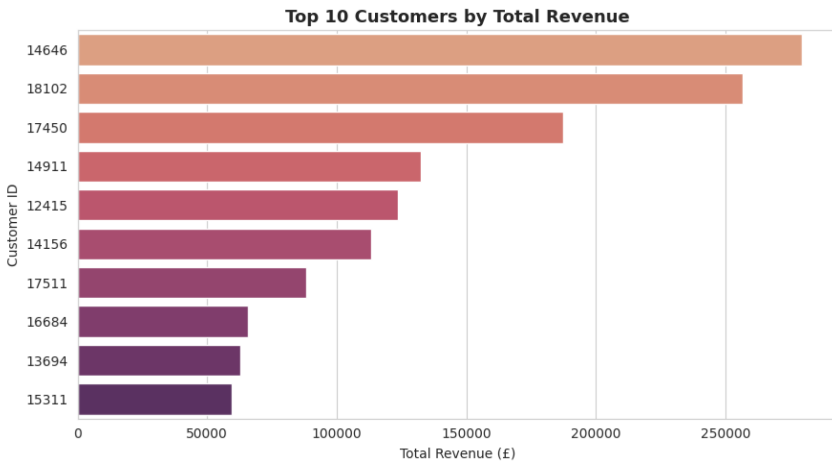

Figure 8: Top 10 customers ranked by total revenue contributed.

The dataset includes 4,372 unique identified customers across 25,900 invoices. Revenue is heavily concentrated among a small group of top-spending customers, suggesting a classic 80/20 customer value distribution.

9. Business Insights

1. **The United Kingdom is the company's largest market.**
Around 84% of the total revenue comes from the UK, making it the company's primary source of sales. Marketing and recommendation efforts should mainly focus on this region.

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2. **A small number of products generate most of the sales.**
Only a few products contribute a significant share of both sales quantity and revenue. These products can be prioritized for promotions and recommendation strategies.

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3. **Sales show clear seasonal trends.**
Revenue increases noticeably during October and November, while the summer months experience lower sales. This pattern can help with inventory planning and marketing campaigns.

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4. **Many transactions have missing customer information.**
About 25% of the records have missing Customer IDs, likely representing guest purchases. This

could affect the performance of a personalized recommendation system and should be addressed before modeling.

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5. **The dataset contains several outliers.**
Extreme values and negative quantities are present due to bulk purchases and cancelled orders. These records should be handled carefully during data preprocessing.

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6. **A small group of customers contributes most of the revenue.**
The top customers generate a large portion of total sales, highlighting the importance of customer retention and loyalty programs.

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7. **Revenue is mainly driven by purchase quantity.**
The strong correlation between Quantity and Revenue suggests that selling more items has a greater impact on revenue than increasing product prices.

10. Conclusion

This exploratory analysis confirms that the dataset is rich enough to support a future recommendation system, but requires a dedicated data-cleaning stage to handle missing CustomerIDs, cancellations, and outliers before modeling. No data was removed at this stage, in line with the task requirements — this report is for observation and pattern discovery only, providing a solid foundation for the next phase of the project.